

I am a professor in automation faculty of a certain university. We have already developed an experimental device for experimental education purposes, consisting of a liquid mixing process consisting of s7-200 PLC, several numerical and analog input/outlet modules (like ADAM 4015, 4017), 20 ambient sensors, temperature sensors, 4 liquid raw material tanks, 6 liquid recovery tanks, 20 electromagnetic valves, a liquid pump, a 6-degree-of-freedom transfer robot, and an automatic warehouse with a loader. As development progresses, we plan to modify this device to allow for practical experience and experimentation with the representation of the Industrial Internet of Things by adding certain sensors, operating devices, and communication devices. Please provide a specific design plan and theoretical basis. Please provide specific information such as the logical structure diagram, communication method, device configuration diagram, data on the required devices, and prices. I am gonna buy the devices in China mainly

Here's a practical, China-buyable **Industrial Internet of Things (IIoT) upgrade design plan** for your experimental liquid mixing device (PLC + sensors + robotic + automation system). I'll cover **design logic**, **communication methods**, **device configuration**, **structure diagrams**, and **approximate device pricing (USD)** from China-friendly sources where available.

1. Theoretical Basis (Why & How IIoT Works Here)

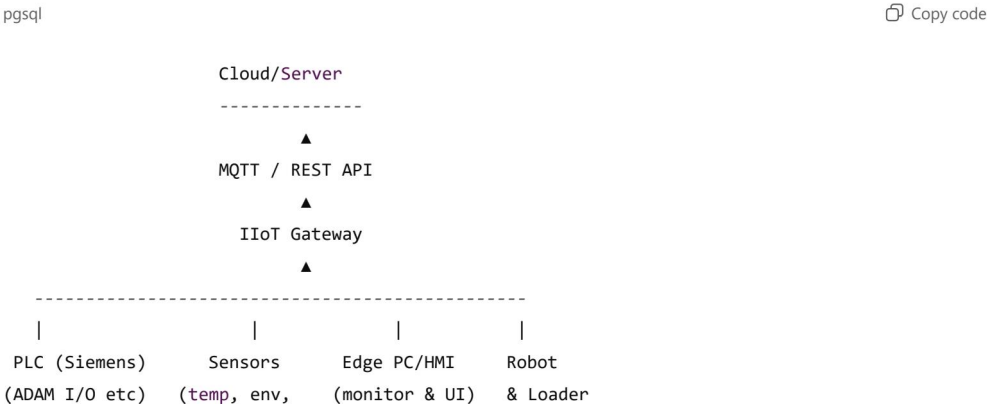
Industrial Internet of Things (IIoT) means networking sensors & devices, letting them send real-time data to a gateway/controller for:

- monitoring and **remote data collection**
- cloud storage/analysis
- advanced control (e.g., predictive maintenance) Wikipedia

In your system:

- **PLC (Siemens S7-200)** remains the real-time controller.
- **IIoT gateways** bridge PLC and sensor network to higher-level systems (Ethernet/Cloud/SCADA).
- Communication uses standards like **Modbus (RTU/TCP)**, **MQTT**, **Ethernet/IP** to connect devices and send data to cloud or local server. Wikipedia

2. Logical Structure Diagram (Simplified)



vibration, etc.)

- PLC reads sensors & controls valves/pump/robot.
- IIoT Gateway connects via **Modbus RS-485 / TCP**.
- Data goes to **cloud or local server**.
- Optional **HMI/SCADA** for visualization.

 3. Communication Methods & Protocols

Layer	Communication	Typical Use
Field	Modbus RTU (RS-485)	Sensor ↔ PLC
Network	Modbus TCP / MQTT / Ethernet/IP	PLC ↔ Gateway/Server
Cloud	MQTT / HTTPS API	Gateway ↔ Cloud

- **Modbus RTU** is simple and industrial-standard. Wikipedia
- **MQTT** is lightweight and suited for many IIoT devices.

 4. Device Configuration (Hardware)

Here are recommended device modules you can integrate:

 IIoT Gateways / Connectivity

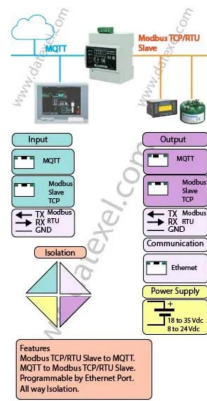
These connect PLC field network to IP/Cloud:



InGateway502 Industrial Cellular Edge Gateway
\$279.00 • InHand Networks



Moxa AIG-302-T Industrial IIoT Gateway
\$927.65 • RSP Supply



Datexel Modbus TCP MQTT Gateway
\$508.00 • Datexel



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\$66

Suggested uses:

- **InGateway502 Industrial Cellular Edge Gateway** – Strong option for edge collection with cellular & cloud support (\$ ~279).
- **Moxa AIG-302-T Industrial IIoT Gateway** – Industrial-grade with expandability (\$ ~928).
- **Datexel Modbus TCP MQTT Gateway** – Good enterprise-level Modbus-to-MQTT (\$ ~508).
- **GRID32 Embedded IIoT Gateway** – Budget embedded gateway (\$ ~66).
- **BI124 IoT Gateway Modbus, EBYTE IoT Gateway Modbus, Serial to Wi-Fi IoT Gateway** – Lower-cost field units (\$15-\$80) for small experiments.
- **LoRaWAN Wireless IO Controller** – If you want wireless sensors in IIoT.

 Sensors (IIoT-Ready or Field Upgrades)

For experiments with added sensor types:

- Temperature / multi-environment (temp / humidity / CO2): [Versatile Outdoor Sensor \(Temp/Humidity/CO2/Light\)](#) (~\$339)
- Environmental: noise sensor: [RS485 Modbus Noise Sensor](#) (~\$139)
- Expand field analog input with low-cost modules: [Low-Cost 2-AI Modbus RS485 ADC Collect](#) (~\$0.99)

For PLC integration you may add:

- **Temperature / Level / Pressure sensors** with 4-20 mA or 0-10 V outputs.
- Wireless options (Bluetooth / LoRa / NB-IoT) for mobile sensor experiments. Made-in-China

 Edge / Visualization Hardware

- A simple industrial **HMI or Edge PC** running SCADA (e.g., Ignition, Node-RED).
- Raspberry Pi / industrial PC can run MQTT brokers and dashboards.

 5. Example Configuration (with Prices)

Component	Qty	Est. Price (USD)	Purpose
IIoT Gateway (edge)	1	\$270-\$928	PLC ↔ Internet/SCADA
Field Gateways (Modbus)	~2	\$15-\$80 each	Connect sensor clusters
Multi-param sensors	~5-10	\$100-\$340	Environmental data
Wireless IO controller	1	\$299	LoRaWAN experiments
Edge PC or HMI	1	\$200-\$600	Visualization
Total: ~ \$1,000–\$3,000 (flexible based on scope)			

 6. Implementation Steps

1. **Network Plan:** Decide wired (Ethernet/RS-485) vs wireless (LoRa/Wi-Fi).
2. **Protocol Mapping:** Ensure sensors/drivers speak **Modbus/RTU/TCP** or use gateways to translate to MQTT.
3. **Gateway Deployment:** Mount IIoT gateway near PLC panel, connect via RS-485 or Ethernet.
4. **Cloud/Server:** Set up MQTT broker (Cloud or local) and database (InfluxDB, SQL).
5. **HMI/SCADA:** Build dashboards showing sensor data and PLC states.
6. **Security:** Use firewalls/VPNs and authenticated protocols (TLS for MQTT).

 7. Conceptual Benefits for Students

- ✓ See **distributed data flow** from sensors to cloud
- ✓ Students learn **industrial protocols (Modbus, MQTT)**
- ✓ Hands-on **edge computing & data visualization**

If you want, I can also draft a **detailed wiring diagram** and a **step-by-step lab exercise plan** based on this architecture.

yes

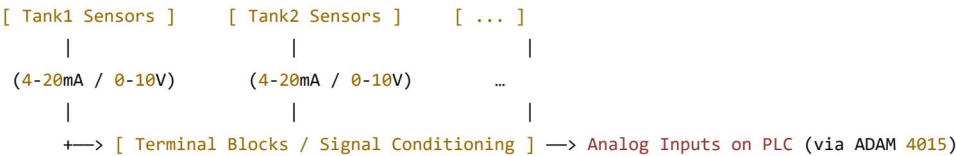
Great — below is a **detailed wiring structure** and a **step-by-step lab exercise plan** you can use with your upgraded IIoT-enabled liquid mixing platform. The diagrams are textual and easy to implement (you can convert them to Visio/AutoCAD later), and the lab plan targets increasing complexity so students learn field wiring, protocols, edge computing, cloud connectivity, and dashboards.

 I. Detailed Wiring & Logical Architecture (Text Diagram)

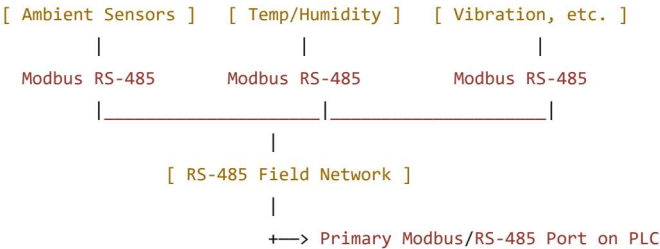
For clarity, I separate this into **Field Wiring (Device Level)**, **Control Network (PLC + Gateways)**, and **Upstream IIoT Network (Edge + Cloud)**.

 A) FIELD-LEVEL WIRING (Sensors, Actuators, PLC)

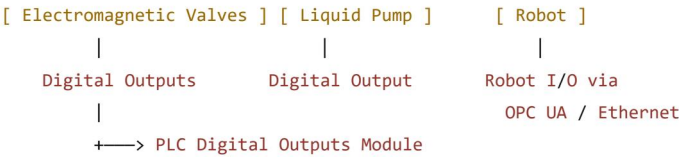
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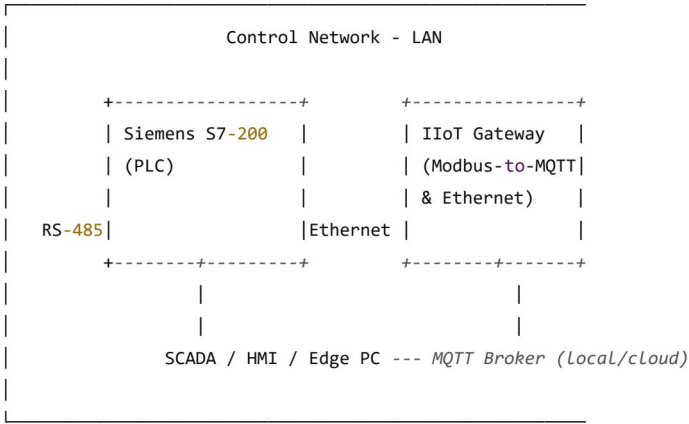


Key Notes

- All **analog sensors** → via ADAM analog module → PLC.
- All **digital actuators** → PLC digital outputs.
- **Modbus RS-485 network** for smart sensors → PLC or IIoT gateway.
- Optional **wireless sensors** (LoRa / NB-IoT) → Wireless IO Gateway → Wired backbone.

 B) PLC + IIoT GATEWAY NETWORK

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Ports & Protocols

- PLC FIELD I/O → sensors & actuators.
- PLC COMM 1: RS-485 (Modbus RTU) → IIoT Gateway (or directly PLC).
- PLC COMM 2: Ethernet (optional) → SCADA / Edge PC.
- IIoT Gateway → MQTT / Modbus TCP (for cloud/SCADA).

C) IIoT NETWORK TO CLOUD (Optional)

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[ IIoT Gateway ] -(Ethernet/Wi-Fi/Cellular)->
    [ Local MQTT Broker ]
        |
        +--> Database (InfluxDB)
        |
        +--> Cloud / Visual Dashboards
```

Protocols

- MQTT/TLS: for secure telemetry.
- REST API: for configuration dashboards.

II. Step-by-Step Lab Exercise Plan

Each exercise can be 90–120 minutes.

Lab 1: Field Wiring & Basic Sensor Integration

Objective

Teach students how to wire sensors and read signals.

Steps

1. Wire a temperature sensor to an analog input module (e.g., ADAM 4015).
2. Configure the PLC ladder to read the analog value.
3. Display the temperature on an HMI.
4. *Assessment:* Show live temperature and inspect wiring integrity.

Learning Outcomes

- Wiring 4–20mA sensors.
- Conditioning analog signals.

- Basic PLC analog scaling.
-

Lab 2: Digital Actuators & Process Control

Objective

Control valves and pumps.

Steps

1. Wire digital outputs to electromagnetic valves & pumps.
2. Create ladder logic to switch valves based on sensor triggers (e.g., level)
3. Debug Valve timing and interlocks.
4. *Assessment:* Automate a fill/empty cycle.

Learner Skills

- Digital I/O wiring.
 - Interlock logic.
 - Safety rules.
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Lab 3: Modbus Device Networking

Objective

Connect a Modbus RS-485 sensor to PLC and/or IIoT gateway.

Steps

1. Connect sensor to RS-485 network.
2. Set parameters: baud rate, parity, address.
3. Read registers in PLC or via gateway.
4. Examine raw registers and convert to engineering values.

Assessment

Working Modbus read cycle and response logging.

Lab 4: IIoT Gateway & Data Publishing

Objective

Configure gateway to publish sensor data to MQTT.

Steps

1. Connect IIoT Gateway to PLC via RS-485.
2. In the gateway web UI: add device, set Modbus parameters.
3. Configure topics for each register → MQTT.
4. Launch local MQTT broker (e.g., Mosquitto).
5. Subscribe to topics and verify messages.

Assessment

Able to see MQTT telemetry on subscriber client.

Lab 5: SCADA/Edge PC Dashboard

Objective

Visualize process data and alarms.

Steps

- 1. Install dashboard software (Node-RED / Ignition / Grafana).
- 2. Connect to the MQTT broker or PLC Ethernet.
- 3. Build real-time visualization of tank levels & valve states.
- 4. Add alarms and historical trending.

Assessment

Working dashboard with historical trend graphs.

 Lab 6: Predictive Maintenance (Optional)

Objective

Use vibration/temp sensors and cloud analysis.

Steps

- 1. Attach machine bearing vibration sensor.
- 2. Publish vibration data via MQTT.
- 3. Use cloud function to calculate trend anomalies.
- 4. Alert when threshold exceeded.

Assessment

Anomaly triggers and maintenance alert.

 Expected Student Skills by End

Skill	Proficiency
Field Wiring	Medium–High
PLC Programming	High
Protocols (Modbus/MQTT)	Medium–High
Gateway Configuration	Medium
Data Visualization	Medium
Cloud Connectivity	Beginner–Intermediate

 Device Price Summary (China-friendly estimates)

Device	Est. Price (USD)
Modbus IIoT Gateway	\$50–\$300
Industrial IIoT Gateway	\$300–\$900
MQTT Broker (Software)	Free
SCADA/HMI Software	\$0–\$\$
Sensors (Temp, CO2, Noise)	\$20–\$350
Wireless IO Gateway (LoRa)	~\$299
Edge PC / HMI	\$200–\$600

Prices approximate, China local sources (AliExpress, Taobao, Made-in-China), quantities affect price.

Tips for Procurement in China

- ✓ Use **Modbus-capable sensors and controllers** to minimize custom drivers.
- ✓ Choose gateways with **SDK support** so your students can build apps.
- ✓ For bulk sensors, use **RS-485 bus topology** to reduce cabling cost.
- ✓ Get devices with **documentation in Chinese** for classroom use.

If you want, I can also generate **PLC ladder logic templates**, **Modbus register maps**, or **dashboard templates** (Grafana/Node-RED) for this system.

